

2.7

SUMMARY

Not all random experiments involve outcomes and events that are numerical. Since engineers and scientists work best with numerical quantities, this chapter is concerned with two things: (1) the definition of a random variable; it is always a numerical quantity, regardless of the random experiment from which it derives; and (2) the various functions that describe the probabilistic behavior of a random variable. Specific principal topics that were covered are:

- The various types of random variables (discrete, continuous, mixed) were defined.
- The concepts of probability density and cumulative probability distribution functions were introduced to define the probabilistic behavior of a random variable.
- The important gaussian random variable was discussed in detail, while others (binomial, Poisson, etc.) were defined.
- Finally, conditional distribution and density functions were discussed to demonstrate how probabilities that are associated with a random variable can depend on some random event.

The chapter's material is basic to understanding topics of the next chapter, which develops ways of operating on (working with) random variables.

PROBLEMS

2.1-1. The sample space for an experiment is $S = \{0, 1, 2.5, 6\}$. List all possible values of the following random variables:

- (a) $X = 2s$
- (b) $X = 5s^2 - 1$
- (c) $X = \cos(\pi s)$
- (d) $X = (1 - 3s)^{-1}$

2.1-2. Work Problem 2.1-1 for $S = \{-2 < s \leq 5\}$.

2.1-3. Given that a random variable X has the following possible values, state if X is discrete, continuous, or mixed.

- (a) $\{-20 < x < -5\}$
- (b) $\{10, 12 < x \leq 14, 15, 17\}$
- (c) $\{-10 \text{ for } s > 2 \text{ and } 5 \text{ for } s \leq 2, \text{ where } 1 < s \leq 6\}$
- (d) $\{4, 3.1, 1, -2\}$

2.1-4. A random variable X is a function. So is probability P . Recall that the *domain* of a function is the set of values its argument may take on while its *range* is the set of corresponding values of the function. In terms of sets, events, and sample spaces, state the domain and range for X and P .

2.1-5. A man matches coin flips with a friend. He wins \$2 if coins match and loses \$2 if they do not match. Sketch a sample space showing possible outcomes for this experiment and illustrate how the points map onto the real line x that

defines the values of the random variable X = "dollars won on a trial." Show a second mapping for a random variable Y = "dollars won by the friend on a trial."

2.1-6. Temperature in a given city varies randomly during any year from -21 to 49°C . A house in the city has a thermostat that assumes only three positions: 1 represents "call for heat below 18.3°C ," 2 represents "dead or idle zone," and 3 represents "call for air conditioning above 21.7°C ." Draw a sample space for this problem showing the mapping necessary to define a random variable X = "thermostat setting."

2.1-7. A random voltage can have any value defined by the set $S = \{a \leq s \leq b\}$. A quantizer divides S into 6 equal-sized contiguous subsets and generates a voltage random variable X having values $\{-4, -2, 0, 2, 4, 6\}$. Each value of X is equal to the midpoint of the subset of S from which it is mapped.

- Sketch the sample space and the mapping to the line x that defines the values of X .
- Find a and b .

***2.1-8.** A random signal can have any voltage value (at a given time) defined by the set $S = \{a_0 < s \leq a_N\}$, where a_0 and a_N are real numbers and N is any integer $N \geq 1$. A voltage quantizer divides S into N equal-sized contiguous subsets and converts the signal level into one of a set of discrete levels a_n , $n = 1, 2, \dots, N$, that correspond to the "input" subsets $\{a_{n-1} < s \leq a_n\}$. The set $\{a_1, a_2, \dots, a_N\}$ can be taken as the discrete values of an "output" random variable X of the quantizer. If the smallest "input" subset is defined by $\Delta = a_1 - a_0$ and other subsets by $a_n - a_{n-1} = 2^{n-1}\Delta$, determine Δ and the quantizer levels a_n in terms of a_0 , a_N , and N .

2.1-9. An honest coin is tossed three times.

- Sketch the applicable sample space S showing all possible elements. Let X be a random variable that has values representing the number of heads obtained on any triple toss. Sketch the mapping of S onto the real line defining X .
- Find the probabilities of the values of X .

2.1-10. Work Problem 2.1-9 for a biased coin for which $P\{\text{head}\} = 0.6$.

2.1-11. Resistor R_2 in Figure P2.1-11 is randomly selected from a box of resistors containing $180\text{-}\Omega$, $470\text{-}\Omega$, $1000\text{-}\Omega$, and $2200\text{-}\Omega$ resistors. All resistor values have the same likelihood of being selected. The voltage E_2 is a discrete random variable. Find the set of values E_2 can have and give their probabilities.

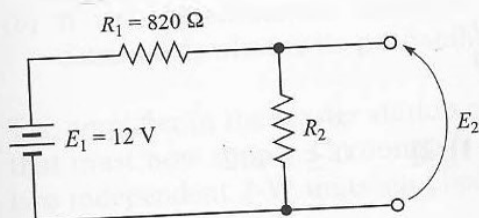


FIGURE P2.1-11

2.1-12. A sample space is defined by $S = \{1, 2 \leq s \leq 3, 4, 5\}$. A random variable is defined by $X = 2$ for $0 \leq s \leq 2.5$, $X = 3$ for $2.5 < s < 3.5$, and $X = 5$ for $3.5 \leq s \leq 6$.

- (a) Is X discrete, continuous, or mixed?
(b) Give a set that defines the values X can have.

2.1-13. A gambler flips a fair coin three times.

- (a) Draw a sample space S for this experiment. A random variable X representing his winnings is defined as follows: He loses \$1 if he gets no heads in three flips; he wins \$1, \$2, and \$3 if he obtains 1, 2, or 3 heads, respectively. Show how elements of S map to values of X .
(b) What are the probabilities of the various values of X ?

2.1-14. A random current is described by the sample space $S = \{-4 \leq i \leq 12\}$. A random variable X is defined by

$$X(i) = \begin{cases} -2 & i \leq -2 \\ i & -2 < i \leq 1 \\ 1 & 1 < i \leq 4 \\ 6 & 4 < i \end{cases}$$

- (a) Show, by a sketch, the value x into which the values of i are mapped by X .
(b) What type of random variable is X ?

2.2-1. Bolts made on a production line are nominally designed to have a 760-mm length. A go-no-go testing device eliminates all bolts less than 650 mm and over 920 mm in length. The surviving bolts are then made available for sale and their lengths are known to be described by a uniform probability density function. A certain buyer orders all bolts that can be produced with a $\pm 5\%$ tolerance about the nominal length. What fraction of the production line's output is he purchasing?

2.2-2. Find and sketch the density and distribution functions for the random variables of parts (a), (b), and (c) in Problem 2.1-1 if the sample space elements have equal likelihoods of occurrence.

2.2-3. If temperature in Problem 2.1-6 is uniformly distributed, sketch the density and distribution functions of the random variable X .

2.2-4. For the uniform random variable defined by (2.5-7) find:

- (a) $P\{0.9a + 0.1b < X \leq 0.7a + 0.3b\}$
(b) $P\{(a + b)/2 < X \leq b\}$

2.2-5. Determine which of the following are valid distribution functions:

- (a) $G_X(x) = \begin{cases} 1 - e^{-x/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$
(b) $G_X(x) = \begin{cases} 0 & x < 0 \\ 0.5 + 0.5 \sin[\pi(x - 1)/2] & 0 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$
(c) $G_X(x) = \frac{x}{a}[u(x - a) - u(x - 2a)]$

2.2-6. A function $G_X(x) = a[1 + (2/\pi) \sin^{-1}(x/c)] \text{rect}(x/2c) + (a+b)u(x-c)$ is defined for all $-\infty < x < \infty$, where $c > 0$, b , and a are real constants and $\text{rect}(\cdot)$ is defined by (E-2). Find any conditions on a , b , and c that will make $G_X(x)$ a valid probability distribution function. Discuss what choices of constants correspond to a continuous, discrete, or mixed random variable.

2.2-7. (a) Generalize Problem 2.2-5(a) by finding values of real constants a and b such that

$$G_X(x) = [1 - a \exp(-x/b)]u(x)$$

is a valid distribution function.

(b) Are there any values of a and b such that $G_X(x)$ corresponds to a mixed random variable X ?

2.2-8. (a) Find the probabilities associated with all values of the random variable X of Problem 2.1-14.

(b) Sketch the probability distribution function of the random variable X .

2.2-9. A random variable X has the distribution function

$$F_X(x) = \sum_{n=1}^{12} \frac{n^2}{650} u(x-n)$$

Find the probabilities: (a) $P\{-\infty < X \leq 6.5\}$, (b) $P\{X > 4\}$, and (c) $P\{6 < X \leq 9\}$.

2.2-10. If the function

$$G_X(x) = K \sum_{n=1}^N n^3 u(x-n)$$

must be a valid probability distribution function, determine K to make it valid. (*Hint:* Use a series from Appendix C.)

2.3-1. Determine the real constant a , for arbitrary real constants m and $0 < b$, such that

$$f_X(x) = ae^{-|x-m|/b}$$

is a valid density function (called the *Laplace density*).

2.3-2. An intercom system master station provides music to six hospital rooms. The probability that any one room will be switched on and draw power at any time is 0.4. When on, a room draws 0.5 W.

(a) Find and plot the density and distribution functions for the random variable "power delivered by the master station."

(b) If the master-station amplifier is overloaded when more than 2 W is demanded, what is its probability of overload?

2.3-3. The amplifier in the master station of Problem 2.3-2 is replaced by a 4-W unit that must now supply 12 rooms. Is the probability of overload better than if two independent 2-W units supplied six rooms each?

2.3-4. Justify that a distribution function $F_X(x)$ satisfies (2.2-2a, b , c).

2.3-5. Use the definition of the impulse function to evaluate the following integrals. (Hint: Refer to Appendix A.)

$$(a) \int_3^4 (3x^2 + 2x - 4)\delta(x - 3.2) dx$$

$$(b) \int_{-\infty}^{\infty} \cos(6\pi x)\delta(x - 1) dx$$

$$(c) \int_{-\infty}^{\infty} \frac{24\delta(x - 2) dx}{x^4 + 3x^2 + 2}$$

$$(d)^\dagger \int_{-\infty}^{\infty} \delta(x - x_0)e^{-j\omega x} dx$$

$$(e) \int_{-3}^3 u(x - 2)\delta(x - 3) dx$$

2.3-6. Show that the properties of a density function $f_X(x)$, as given by (2.3-6), are valid.

2.3-7. For the random variable defined in Example 2.3-1, find:

- (a) $P\{x_0 - 0.6\alpha < X \leq x_0 + 0.3\alpha\}$
 (b) $P\{X = x_0\}$

2.3-8. Find a constant $b > 0$ so that the function

$$f_X(x) = \begin{cases} e^{3x}/4 & 0 \leq x \leq b \\ 0 & \text{elsewhere} \end{cases}$$

is a valid probability density.

2.3-9. Given the function

$$g_X(x) = 4 \cos(\pi x/2b) \text{rect}(x/2b)$$

find a value of b so that $g_X(x)$ is a valid probability density.

2.3-10. A random variable X has the density function

$$f_X(x) = \left(\frac{1}{2}\right)u(x) \exp(-x/2)$$

Define events $A = \{1 < X \leq 3\}$, $B = \{X \leq 2.5\}$, and $C = A \cap B$. Find the probabilities of events (a) A , (b) B , and (c) C .

***2.3-11.** Let $\phi(x)$ be a continuous, but otherwise arbitrary real function, and let a and b be real constants. Find $G(a, b)$ defined by

$$G(a, b) = \int_{-\infty}^{\infty} \phi(x) \delta(ax + b) dx$$

(Hint: Use the definition of the impulse function.)

[†]The quantity j is the unit-imaginary; that is, $j = \sqrt{-1}$.

- 2.3-12.** For real constants $b > 0$, $c > 0$, and any a , find a condition on constant a and a relationship between c and a (for given b) such that the function

$$f_X(x) = \begin{cases} a[1 - (x/b)] & 0 \leq x \leq c \\ 0 & \text{elsewhere} \end{cases}$$

is a valid probability density.

- 2.3-13.** Use the properties or definition of the impulse function (Appendix A) to evaluate the following integrals:

(a) $\int_{-\infty}^{\infty} \delta(x+5) \frac{x^2}{1+x^2} dx$

(b) $\int_{-\infty}^{\infty} \delta(x-3) \cos(\pi x/6) dx$

(c) $\int_{-\infty}^{\infty} e^{-4(x+1)} \delta(x+1) dx$

- 2.3-14.** Work Problem 2.3-13 except for the following integrals:

(a) $\int_{-2}^6 [\delta(x-1) + \delta(x+3) + \delta(x-5)] dx$

(b) $\int_{-\infty}^6 \delta(x-7) u(x+3) dx$

(c) $\int_{-3}^2 [\delta(x-1) - \delta(x+2)] \frac{e^{-2x^2}}{1+x^2+x^4} dx$

- 2.3-15.** Find a value for constant A such that

$$f_X(x) = \begin{cases} 0 & x < -1 \\ A(1-x^2) \cos(\pi x/2) & -1 \leq x \leq 1 \\ 0 & 1 < x \end{cases}$$

is a valid probability density function.

- 2.4-1.** A random variable X is gaussian with $a_X = 0$ and $\sigma_X = 1$.

- (a) What is the probability that $|X| > 2$?
 (b) What is the probability that $X > 2$?

- 2.4-2.** Work Problem 2.4-1 if $a_X = 4$ and $\sigma_X = 2$.

- 2.4-3.** For the gaussian density function of (2.4-1), show that

$$\int_{-\infty}^{\infty} x f_X(x) dx = a_X$$

- 2.4-4.** For the gaussian density function of (2.4-1), show that

$$\int_{-\infty}^{\infty} (x - a_X)^2 f_X(x) dx = \sigma_X^2$$

- 2.4-5.** A production line manufactures 1000- Ω resistors that must satisfy a 10% tolerance.

- (a) If resistance is adequately described by a gaussian random variable X for which $a_X = 1000 \Omega$ and $\sigma_X = 40 \Omega$, what fraction of the resistors is expected to be rejected?
- (b) If a machine is not properly adjusted, the product resistances change to the case where $a_X = 1050 \Omega$ (5% shift). What fraction is now rejected?
- 2.4-6.** Cannon shell impact position, as measured along the line of fire from the target point, can be described by a gaussian random variable X . It is found that 15.15% of shells fall 11.2 m or farther from the target in a direction toward the cannon, while 5.05% fall farther than 95.6 m beyond the target. what are a_X and σ_X for X ?
- 2.4-7.** A gaussian random variable X has $a_X = 2$, and $\sigma_X = 2$.
- (a) Find $P\{X > 1.0\}$.
- (b) Find $P\{X \leq -1.0\}$.
- 2.4-8.** In a certain "junior" olympics, javelin throw distances are well approximated by a gaussian distribution for which $a_X = 30$ m and $\sigma_X = 5$ m. In a qualifying round, contestants must throw farther than 26 m to qualify. In the main event the record throw is 42 m.
- (a) What is the probability of being disqualified in the qualifying round?
- (b) In the main event what is the probability the record will be broken?
- 2.4-9.** Suppose height to the bottom of clouds is a gaussian random variable X for which $a_X = 4000$ m, and $\sigma_X = 1000$ m. A person bets that cloud height tomorrow will fall in the set $A = \{1000 \text{ m} < X \leq 3300 \text{ m}\}$ while a second person bets that height will be satisfied by $B = \{2000 \text{ m} < X \leq 4200 \text{ m}\}$. A third person bets they are both correct. Find the probabilities that each person will win the bet.
- 2.4-10.** The output voltage X from the receiver in a particular binary digital communication system, when a binary zero is being received, is gaussian (noise only) as defined by $a_X = 0$ and $\sigma_X = 0.3$. When a binary one is being received it is also gaussian (signal-plus-noise now), but as defined by $a_X = 0.9$ and $\sigma_X = 0.3$. The receiver's decision logic specifies that at the end of a binary (bit) interval, if $X > 0.45$ a binary one is being received. If $X \leq 0.45$ a binary zero is decided. If it is given that a binary zero is truly being received, find the probabilities that (a) a binary one (mistake) will be decided, and (b) a binary zero is decided (correct decision).
- 2.4-11.** A gaussian random voltage X for which $a_X = 0$ and $\sigma_X = 4.2$ V appears across a 100- Ω resistor with a power rating of 0.25 W. What is the probability that the voltage will cause an instantaneous power that exceeds the resistor's rating?
- 2.4-12.** Work Problem 2.4-11 except assume a 0.5-W resistor.
- 2.4-13.** For the gaussian random variable, show that the curve's points of inflection (where the first derivative of the probability density function with respect to X has a zero slope) occur at $a_X \pm \sigma_X$.

2.4-14. A random variable X is known to be gaussian with $a_X = 1.6$ and $\sigma_X = 0.4$. Find: (a) $P\{1.4 < X \leq 2.0\}$, and (b) $P\{-0.6 < (X - 1.6) \leq 0.6\}$.

2.4-15. The radial distance to the impact points for shells fired over land by a cannon is well-approximated as a gaussian random variable with $a_X = 2000$ m and $\sigma_X = 40$ m when the cannon is aimed at a target located at 1980 m distance.
(a) Find the probability that shells will fall within ± 68 m of the target.
(b) Find the probability that shells will fall at distances of 2050 m or more.

2.4-16. Assume that the time of arrival of birds at a particular place on a migratory route, as measured in days from the first of the year (January 1 is the first day), is approximated as a gaussian random variable X with $a_X = 200$ and $\sigma_X = 20$ days.
(a) What is the probability the birds arrive after 160 days but on or before the 210th day?
(b) What is the probability the birds will arrive after the 231st day?

2.4-17. Assume fluorescent lamps made by a manufacturer have a probability of 0.05 of being inoperable when new. A person purchases eight of the lamps for home use.
(a) Plot the probability distribution function for a random variable "the number of inoperable lamps."
(b) What is the probability that exactly one lamp is inoperable of the eight?
(c) What is the probability that all eight lamps are functional?
(d) Determine the probability that one or more lamps are not operable.

2.5-1. (a) Use the exponential density of (2.5-9) and solve for I_2 defined by

$$I_2 = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

(b) Solve for I_1 defined by

$$I_1 = \int_{-\infty}^{\infty} x f_X(x) dx$$

(c) Verify that I_1 and I_2 satisfy the equation $I_2 - I_1^2 = b^2$.

2.5-2. Verify that the maximum value of $f_X(x)$ for the Rayleigh density function of (2.5-11) occurs at $x = a + \sqrt{b/2}$ and is equal to $\sqrt{2/b} \exp(-\frac{1}{2}) \approx 0.607\sqrt{2/b}$. This value of x is called the *mode* of the random variable. (In general, a random variable may have more than one such value—explain.)

2.5-3. The lifetime of a system expressed in weeks is a Rayleigh random variable X for which

$$f_X(x) = \begin{cases} (x/200)e^{-x^2/400} & 0 \leq x \\ 0 & x < 0 \end{cases}$$

(a) What is the probability that the system will not last a full week?
(b) What is the probability the system lifetime will exceed one year?

2.5-4. The *Cauchy*[†] random variable has the probability density function

$$f_X(x) = \frac{b/\pi}{b^2 + (x - a)^2}$$

for real numbers $0 < b$ and $-\infty < a < \infty$. Show that the distribution function of X is

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1} \left(\frac{x - a}{b} \right)$$

2.5-5. The *log-normal density* function is given by

$$f_X(x) = \begin{cases} \frac{\exp\{-[\ln(x - b) - a_X]^2 / 2\sigma_X^2\}}{\sqrt{2\pi}\sigma_X(x - b)} & x \geq b \\ 0 & x < b \end{cases}$$

for real constants $0 < \sigma_X$, $-\infty < a_X < \infty$, and $-\infty < b < \infty$, where $\ln(x)$ denotes the natural logarithm of x . Show that the corresponding distribution function is

$$F_X(x) = \begin{cases} F\left[\frac{\ln(x - b) - a_X}{\sigma_X}\right] & x \geq b \\ 0 & x < b \end{cases}$$

where $F(\cdot)$ is given by (2.4-3).

2.5-6. A random variable X is known to be Poisson with $b = 4$.

- Plot the density and distribution functions for this random variable.
- What is the probability of the event $\{0 \leq X \leq 5\}$?

2.5-7. The number of cars arriving at a certain bank drive-in window during any 10-min period is a Poisson random variable X with $b = 2$. Find:

- The probability that more than 3 cars will arrive during any 10-min period.
- The probability that no cars will arrive.

2.5-8. Let X be a Rayleigh random variable with $a = 0$. Find the probability that X will have values larger than its mode (see Problem 2.5-2).

2.5-9. A certain large city averages three murders per week and their occurrences follow a Poisson distribution.

- What is the probability that there will be five or more murders in a given week?
- On the average, how many weeks a year can this city expect to have no murders?
- How many weeks per year (average) can the city expect the number of murders per week to equal or exceed the average number per week?

2.5-10. A certain military radar is set up at a remote site with no repair facilities. If the radar is known to have a *mean-time-between-failures* (MTBF) of 200 h, find

[†]After the French mathematician Augustin Louis Cauchy (1789–1857).

the probability that the radar is still in operation one week later when picked up for maintenance and repairs.

2.5-11. If the radar of Problem 2.5-10 is permanently located at the remote site, find the probability that it will be operational as a function of time since its setup.

2.5-12. A computer undergoes downtime if a certain critical component fails. This component is known to fail at an average rate of once per four weeks. No significant downtime occurs if replacement components are on hand because repair can be made rapidly. There are three components on hand, and ordered replacements are not due for six weeks.

(a) What is the probability of significant downtime occurring before the ordered components arrive?

(b) If the shipment is delayed two weeks, what is the probability of significant downtime occurring before the shipment arrives?

2.5-13. The envelope (amplitude) of the output signal of a radar system that is receiving only noise (no signal) is a Rayleigh random voltage X for which $a = 0$ and $b = 2$ V. The system gets a false target detection if X exceeds a threshold level V volts. How large must V be to make the probability of false detection 0.001?

2.6-1. Rework Example 2.6-1 to find $f_X(x|B_2)$ and $F_X(x|B_2)$. Sketch the two functions.

2.6-2. Extend the analysis of the text that leads to (2.6-11) and (2.6-12) to the more general event $B = \{a < X \leq b\}$. Specifically, show that now

$$F_X(x|a < X \leq b) = \begin{cases} 0 & x < a \\ \frac{F_X(x) - F_X(a)}{F_X(b) - F_X(a)} & a \leq x < b \\ 1 & b \leq x \end{cases}$$

and

$$f_X(x|a < X \leq b) = \begin{cases} 0 & x < a \\ \frac{f_X(x)}{F_X(b) - F_X(a)} = \frac{f_X(x)}{\int_a^b f_X(x) dx} & a \leq x < b \\ 0 & b \leq x \end{cases}$$

2.6-3. Consider the system having a lifetime defined by the random variable X in Problem 2.5-3. Given that the system will survive beyond 20 weeks, find the probability that it will survive beyond 26 weeks.

2.6-4. Assume the lifetime of a laboratory research animal is defined by a Rayleigh density with $a = 0$ and $b = 30$ weeks in (2.5-11) and (2.5-12). If for some clinical reasons it is known that the animal will live at *most* 20 weeks, what is the probability it will live 10 weeks or less?

2.6-5. Suppose the depth of water, measured in meters, behind a dam is described by an exponential random variable having a density

$$f_X(x) = (1/13.5)u(x) \exp(-x/13.5)$$

There is an emergency overflow at the top of the dam that prevents the depth from exceeding 40.6 m. There is a pipe placed 32.0 m below the overflow (ignore the pipe's finite diameter) that feeds water to a hydroelectric generator.

- What is the probability that water is wasted through emergency overflow?
- Given that water is not wasted in overflow, what is the probability that the generator will have water to drive it?
- What is the probability that water will be too low to produce power?

***2.6-6.** In Problem 2.6-5 find and sketch the distribution and density functions of water depth given that water will be deep enough to generate power but no water is wasted by emergency overflow. Also sketch for comparison the distribution and density of water depth without any conditions.

***2.6-7.** In Example 2.6-2 a parachuter is an "expert" if he hits the bull's eye. If he falls outside the bull's eye but within a circle of 25-m radius, he is called "qualified for competition." Given that a parachuter is not an expert but hits the target, what is the probability of being "qualified"?

2.6-8. In a game show contestants choose one of three doors to determine what prize they win. History shows that the three doors, 1, 2, and 3, are chosen with probabilities 0.30, 0.45, and 0.25, respectively. It is also known that given door 1 is chosen, the probabilities of winning prizes of \$0, \$100, and \$1000 are 0.10, 0.20, and 0.70. For door 2 the respective probabilities are 0.50, 0.35, and 0.15, and for door 3 they are 0.80, 0.15, and 0.05. If X is a random variable describing dollars won, and D describes the door selected (values of D are $D_1 = 1$, $D_2 = 2$, and $D_3 = 3$), find: (a) $F_X(x|D = D_1)$ and $f_X(x|D = D_1)$, (b) $f_X(x|D = D_2)$, (c) $f_X(x|D = D_3)$, and (d) $f_X(x)$.

2.6-9. Again consider the game show of Problem 2.6-8 and find the probabilities of winning (a) \$0, (b) \$100, and (c) \$1000.

***2.6-10.** Divers return each day to the site of a sunken treasure ship. Due to random navigational errors, they arrive with a radial positional error (from the true site) described by a random variable X (in kilometers) defined by

$$F_X(x) = [1 - e^{-x^3/2}]u(x)$$

(this is the *Weibull*[†] distribution; see Appendix F). If they must arrive with an error of not more than 1.2 km to prevent having to move to a new position, and within 0.6 km for optimum use of air tanks, what is the probability of optimum use of tanks given they arrive on site on the first effort?

***2.6-11.** For the navigational errors of Problem 2.6-10 find and plot the conditional density $f_X(x|X \leq 1.2 \text{ km})$.

[†]After Ernst Hjalmar Waloddi Weibull (1887–1979), a Swedish applied physicist.